Minimum Wages and Fast Food Employment

An AI Assisted Empirical Replication of Card and Krueger 1994

Individual project report | Applied labor economics | Difference in differences

Project Overview

This is an AI-assisted empirical paper replication project in applied labor economics. It reconstructs the data, variables, difference-in-differences design, assigned tables, and figure from Card and Krueger's 1994 study of the New Jersey minimum-wage increase. The project then adds an independent extension testing whether the employment response differs between company-owned and franchise restaurants.

The central finding is that the supplied public data reproduce the paper's headline result: New Jersey fast-food employment increased relative to eastern Pennsylvania after the policy change, rather than declining. The AI-assisted workflow does not replace researcher responsibility for data choices, validation, interpretation, citations, or final submission.

1 Research Question and Economic Mechanism

The study asks whether New Jersey's April 1, 1992 increase in the minimum wage from $4.25 to $5.05 reduced employment at fast-food restaurants. A higher wage floor can reduce labor demand through higher costs, but firms may also adjust through prices, turnover, hours, effort, or other margins. In a labor market with employer market power, a higher wage could even increase employment. The empirical question is therefore open and must be evaluated using a credible comparison group.

2 Data and Variable Construction

The supplied public file contains 410 restaurants: 331 in New Jersey and 79 in Pennsylvania. Restaurants were surveyed before the policy change in February-March 1992 and again in November-December 1992. The assigned replication covers Tables 2-4 and Figure 1; restricted data for Table 1 are outside the replication scope.

Full-time-equivalent employment is defined as EMPFT + 0.5 x EMPPT + NMGRS. Permanent closures are coded as zero post-period employment. Temporary, renovation, highway-construction, and mall-fire closures are missing in the main post-period outcome and are set to zero only in the explicit closure-coding sensitivity check.

The Table 4 sample requires observed pre- and post-period FTE employment, baseline wage, and post-period wage unless the restaurant permanently closed. This produces 357 observations. The treatment-intensity variable is GAP = max(0, (5.05 - WAGE_ST) / WAGE_ST) for New Jersey and zero for Pennsylvania.

3 Empirical Design

The core difference-in-differences estimate is (NJ after - NJ before) - (PA after - PA before). New Jersey is the treatment group, and eastern Pennsylvania is the comparison group. The identifying assumption is that, absent the policy, employment changes would have been parallel across the two regions. Geographic proximity supports the comparison, but the data contain only one pre-treatment survey wave, so parallel trends cannot be directly tested with an event-study design.

4 Core Replication Results

The unbalanced Table 3 difference-in-differences estimate is 2.75 FTE employees with a standard error of 1.34, compared with the published value of 2.76. The balanced-sample estimate is 2.75, and the temporary-closures-as-zero sensitivity estimate is 2.51. The direction and magnitude are consistent with the paper's conclusion that the policy did not reduce fast-food employment relative to the comparison group.

Table 1. Headline replication estimates

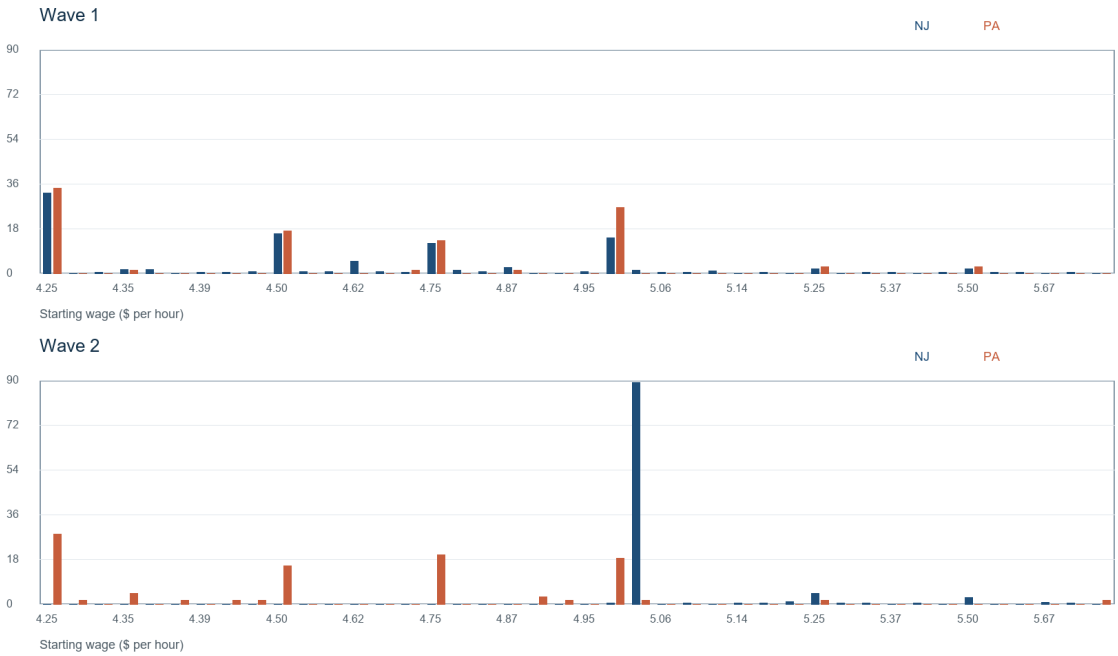
Result	Replicated	SE	Original
Table 3 unbalanced DiD	2.75	1.34	2.76
Table 3 balanced DiD	2.75		2.75
Table 3 temporary closures as zero	2.51		2.51

The Table 4 NJ-indicator coefficient is 2.33 with a standard error of 1.19. The continuous GAP specification estimates 15.65 with a standard error of 6.08. The selected published-versus-replicated comparisons are close; the full comparison file is included in the reproducibility package.

Table 2. Published-versus-replicated results

result	original_paper	replicated	difference
Table 3 row 3 DiD	2.76	2.75	-0.01
Table 3 row 4 DiD	2.75	2.75	0.00
Table 3 row 5 DiD	2.51	2.51	-0.00
Table 4 (i) NJ dummy STATE	2.33	2.33	-0.00
Table 4 (ii) NJ dummy + chain/ownership STATE	2.30	2.30	0.00
Table 4 (iii) GAP GAP	15.65	15.65	0.00
Table 4 (iv) GAP + chain/ownership GAP	14.92	14.92	-0.00
Table 4 (v) GAP + chain/ownership + region GAP	11.91	11.98	0.07

Figure 1. Distribution of starting wage rates



Notes: Bars show the percentage of restaurants at each observed starting wage in New Jersey and Pennsylvania.

5 Independent Extension Ownership Heterogeneity

The extension asks whether the employment response differs between company-owned and franchise stores. The franchise-store GAP coefficient is 17.08; the GAP x company-owned interaction is -14.46; and the implied company-owned marginal effect is 2.61. These point estimates suggest a possible ownership difference, but the interaction is imprecisely estimated. The result should therefore be presented as exploratory evidence, not as a definitive heterogeneous treatment effect.

Table 3. Ownership extension estimates

term	quantity	reported_estimate	standard_error
constant	coefficient	-0.59	2.06
BK	coefficient	-0.17	1.53
KFC	coefficient	0.38	1.72
roys	coefficient	-2.24	1.73
CO_OWNED	coefficient	1.35	1.54
SOUTHJ	coefficient	-0.23	1.92
CENTRALJ	coefficient	-1.56	2.01
NORTHJ	coefficient	-0.16	1.75
PA1	coefficient	-3.51	2.24
GAP	coefficient	17.08	8.72
GAP_CO_OWNED	coefficient	-14.46	12.99
marginal_effect	GAP effect, franchise stores	17.08	8.72
marginal_effect	GAP effect, company-owned stores	2.61	11.21

6 Robustness and Validation

The workflow tests alternative part-time weights of 0.4 and 0.6, excludes Shore stores, restricts the sample to stores with at most two first-wave call attempts, removes each chain in turn, changes temporary-closure coding, and performs 2,000 state-label permutations while preserving the number of New Jersey observations.

Table 4. Robustness checks

Check	Specification	N	Estimate	SE	Permutation p
Baseline	NJ dummy; published Table 4 model (i)	357	2.33	1.19	
Alternative part-time weight	NJ dummy; part-time weight=0.4	357	2.37	1.20	
Alternative part-time weight	NJ dummy; part-time weight=0.6	357	2.28	1.20	
Geographic exclusion	NJ dummy; exclude Shore stores	326	2.56	1.19	
Survey follow-up exclusion	NJ dummy; NCALLS <= 2	294	2.45	1.28	
Leave-one-chain-out	NJ dummy; exclude Burger King	209	1.51	1.36	
Leave-one-chain-out	NJ dummy; exclude KFC	283	3.06	1.38	
Leave-one-chain-out	NJ dummy; exclude Roy Rogers	267	2.42	1.43	
Leave-one-chain-out	NJ dummy; exclude Wendy's	312	2.05	1.29	
Closure coding	Difference-in-differences; temporary closures set to zero	388	2.51	1.35	
Randomization inference	NJ dummy; 2,000 state-label permutations	357	2.33	1.19	0.05

The NJ-indicator estimate remains positive under the alternative FTE weights and sample exclusions. The permutation p-value for the baseline coefficient is approximately 0.050. These checks are diagnostics around the baseline design and do not eliminate the need to defend the parallel-trends assumption.

7 Limitations

- Only one pre-treatment wave is available, limiting direct assessment of parallel trends.
- Employment is survey-reported and may contain measurement error.
- Closure coding and missing post-period values affect the estimand, so the coding rule must be stated clearly.
- The sample covers four fast-food chains and two neighboring regions in 1992; external validity is limited.
- The definitions of some regional indicators in the supplied data dictionary require final confirmation against the original source codebook.

8 Conclusion

The replication recovers the central Card-Krueger result: the New Jersey minimum-wage increase is not associated with a decline in fast-food employment relative to eastern Pennsylvania in the supplied public data. The estimate is stable across the main closure and sample checks. The ownership extension is economically motivated but should remain clearly labelled as exploratory.

References

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." American Economic Review 84(4): 772-793.

Course-provided public replication data and data dictionary. Access instructions are documented in the project README and data/raw/README.md.

The project uses an AI-assisted workflow. The specific uses, checks, corrections, and remaining researcher responsibilities are documented in docs/ai_use_disclosure.md.

Appendix A Assigned Table Outputs

The machine-readable CSV files in outputs/tables are the authoritative generated outputs. The compact tables below provide report-readable versions; the complete CSV files preserve all rows and columns.

Appendix Table A1. Table 2 descriptive statistics

section	variable	NJ	PA	t_NJ_minus_PA	n_NJ	n_PA
Store types	Burger King	41.09	44.30	-0.52	331	79
Store types	KFC	20.54	15.19	1.16	331	79
Store types	Roy Rogers	24.77	21.52	0.62	331	79
Store types	Wendy's	13.60	18.99	-1.12	331	79
Store types	Company-owned	34.14	35.44	-0.22	331	79
Wave 1	FTE employment	20.44	23.33	-2.00	321	77
Wave 1	Percentage full-time employees	32.85	35.04	-0.72	321	77
Wave 1	Starting wage	4.61	4.63	-0.40	314	76
Wave 1	Wage = $4.25	30.51	32.91	-0.41	331	79
Wave 1	Price of full meal	3.35	3.04	3.96	311	76
Wave 1	Hours open	14.42	14.53	-0.29	331	79
Wave 1	Recruiting bonus	23.56	29.11	-0.98	331	79
Wave 2	FTE employment	21.03	21.17	-0.13	319	77
Wave 2	Percentage full-time employees	35.87	30.38	1.75	314	76
Wave 2	Starting wage	5.08	4.62	10.82	318	71
Wave 2	Wage = $4.25	0.00	25.32	-5.14	331	79
Wave 2	Wage = $5.05	85.50	1.27	36.38	331	79
Wave 2	Price of full meal	3.41	3.03	5.08	305	71
Wave 2	Hours open	14.42	14.65	-0.65	321	78
Wave 2	Recruiting bonus	20.32	23.38	-0.57	315	77

Appendix Table A2. Table 3 employment changes

row	stat	PA	NJ	NJ-PA	NJ low	NJ mid	NJ high	low-high	mid-high
1. FTE employment before	estimate	23.33	20.44	-2.89	19.56	20.08	22.25	-2.69	-2.17
1. FTE employment before	standard_error	1.35	0.51	1.44	0.77	0.84	1.14	1.37	1.41
2. FTE employment after	estimate	21.17	21.03	-0.14	20.88	20.96	20.21	0.66	0.74
2. FTE employment after	standard_error	0.94	0.52	1.08	1.01	0.76	1.03	1.44	1.27
3. Change in mean FTE	estimate	-2.28	0.47	2.75	1.20	0.71	-2.16	3.36	2.87
3. Change in mean FTE	standard_error	1.25	0.48	1.34	0.82	0.69	1.01	1.30	1.22
4. Change in mean FTE, balanced sample	estimate	-2.28	0.47	2.75	1.20	0.71	-2.16	3.36	2.87
4. Change in mean FTE, balanced sample	standard_error	1.25	0.48	1.34	0.82	0.69	1.01	1.30	1.22
5. Change in mean FTE, temporary closures set to zero	estimate	-2.28	0.23	2.51	0.90	0.49	-2.39	3.29	2.88
5. Change in mean FTE, temporary closures set to zero	standard_error	1.25	0.49	1.35	0.87	0.69	1.02	1.34	1.23

Appendix Table A3. Table 4 key coefficients

model	term	coefficient	standard_error	t_stat	N	SER	R_squared
(i) NJ dummy	STATE	2.33	1.19	1.95	357	8.79	0.01
(ii) NJ dummy + chain/ownership	STATE	2.30	1.20	1.93	357	8.78	0.02
(iii) GAP	GAP	15.65	6.08	2.57	357	8.76	0.02
(iv) GAP + chain/ownership	GAP	14.92	6.21	2.40	357	8.76	0.03
(v) GAP + chain/ownership + region	GAP	11.98	7.42	1.61	357	8.75	0.04